Supplementary Materials

For the article “An eye tracking study of textbook reading: The impact of popularisation techniques on interest and processing”

Notation. M (SD) - mean (standard deviation). b - unstandardized coefficient of a linear mixed-effects model; SE - its standard error. Models were fitted in R using the lme4 and lmerTest packages. The between-person component of a predictor is the participant’s mean relative to the sample mean; the within-person component is the deviation of a given text's value from the participant’s mean.

Texts. “Market” and “Social norms” - social studies; “Relief” and “Soil” - geography. Each text was presented in either the original or the modified version.

List of tables

Table S1. Eye-movement measures at the text level by version
Table S2. Eye-movement measures at the text level by text and version
Table S3. Eye-movement measures in paragraphs with modification techniques
Table S4. Eye-movement measures on unchanged text fragments
Table S5. Participants’ prior knowledge by text topic
Table S6. Post-reading text ratings by text and version
Table S7. Text comprehension by text and version
Table S8. Percentage of correct answers to individual comprehension questions
Table S9. Effect of text version on eye-movement measures at the text level
Table S10. Effect of text version on eye-movement measures in first paragraphs (contextualization)
Table S11. Effect of text version on eye-movement measures in second and third paragraphs (dialogization and cohesion)
Table S12. Effect of text version on reading of unchanged fragments
Table S13. Effect of text version on post-reading ratings
Table S14. Effect of text version on comprehension
Table S15. Prior knowledge and eye-movement measures at the text level
Table S16. Correlations of prior knowledge with post-reading ratings
Table S17. Correlations of prior knowledge with text comprehension
Table S18. Post-reading ratings and eye-movement measures at the text level
Table S19. Correlations of post-reading ratings with eye-movement measures
Table S20. Correlations of post-reading ratings with comprehension
Table S21. Comprehension and eye-movement measures at the text level

Descriptive statistics

Table S1. Eye-movement measures at the text level by version

Measure	Original version	Modified version
Reading time per character, ms	63.5 (22.7)	58.0 (18.0)
Reading time per word, ms	415 (144)	346 (112)
Fixations per character	0.257 (0.078)	0.238 (0.068)
Fixations per word	1.68 (0.49)	1.42 (0.42)
Mean fixation duration, ms	245 (28)	243 (26)
Forward saccade amplitude, deg	7.19 (1.57)	7.26 (1.75)
Regressive saccade amplitude, deg	5.55 (1.52)	5.54 (1.36)
Proportion of regressive saccades	0.557 (0.185)	0.552 (0.169)

Note. Values are M (SD) across all texts.

Table S2. Eye-movement measures at the text level by text and version

	"Market"		"Social norms"		"Relief"		"Soil"	
Measure	Orig.	Mod.	Orig.	Mod.	Orig.	Mod.	Orig.	Mod.
Reading time per character, ms	59.6 (19.7)	54.1 (12.5)	56.0 (12.6)	54.4 (17.5)	73.9 (27.9)	58.8 (15.8)	62.8 (23.6)	64.6 (22.9)
Reading time per word, ms	378 (125)	305 (71)	392 (88)	332 (107)	461 (174)	339 (91)	425 (160)	402 (143)
Fixations per character	0.250 (0.07)	0.226 (0.05)	0.230 (0.05)	0.226 (0.06)	0.295 (0.09)	0.237 (0.07)	0.247 (0.08)	0.261 (0.08)
Fixations per word	1.59 (0.45)	1.27 (0.29)	1.61 (0.35)	1.38 (0.39)	1.84 (0.57)	1.37 (0.38)	1.67 (0.54)	1.62 (0.52)
Mean fixation duration, ms	236 (32)	240 (19)	244 (20)	238 (30)	247 (33)	249 (25)	252 (24)	245 (30)
Forward saccade amplitude, deg	7.38 (1.92)	7.38 (1.53)	7.26 (1.14)	7.81 (2.20)	7.32 (1.59)	6.94 (1.18)	6.80 (1.53)	6.89 (1.81)
Regressive saccade amplitude, deg	5.24 (1.14)	5.60 (1.25)	6.23 (1.60)	5.65 (1.45)	5.21 (1.20)	6.03 (1.56)	5.65 (1.92)	4.96 (1.03)
Proportion of regressive saccades	0.609 (0.17)	0.504 (0.16)	0.541 (0.16)	0.608 (0.17)	0.627 (0.19)	0.530 (0.19)	0.445 (0.16)	0.559 (0.16)

Note. Values are M (SD). Orig. — original version; Mod. — modified version.

Table S3. Eye-movement measures in paragraphs with modification techniques

Measure	Original version	Modified version
Contextualization (first paragraph)		
Reading time per character, ms	71.9 (35.7)	60.4 (19.6)
Fixations per character	0.293 (0.127)	0.249 (0.070)
Mean fixation duration, ms	241 (32)	240 (28)
Dialogization		

Measure	Original version	Modified version
Reading time per character, ms	61.7 (21.6)	57.0 (20.6)
Fixations per character	0.250 (0.076)	0.234 (0.080)
Mean fixation duration, ms	245 (29)	243 (28)
Cohesion		
Reading time per character, ms	61.3 (25.8)	58.4 (23.3)
Fixations per character	0.248 (0.094)	0.239 (0.090)
Mean fixation duration, ms	245 (30)	243 (33)

Note. Values are M (SD). For the original version, the corresponding paragraph without the technique.

Table S4. Eye-movement measures on unchanged text fragments

Measure	Original version	Modified version
Fragment reading time, ms	11,446 (6727)	10,152 (6003)
Fixations on fragment	46.1 (25.0)	41.4 (23.9)
Mean fixation duration, ms	245 (33)	244 (32)

Note. Values are M (SD). Unchanged fragments are identical in both versions.

Table S5. Participants' prior knowledge by text topic

		Answer accuracy, %			Heard of the topic, %	Confidence, M (SD)	Studied the topic at school, %		
Topic	n	Correct	Partially correct	Incorrect			Yes	No	Don't remember
Market	43	63	26	12	84	3.02 (1.12)	53	12	35
Social norms	43	58	21	21	95	3.37 (1.09)	63	9	28
Relief	43	56	35	9	58	2.49 (1.01)	42	12	47
Soil	43	65	30	5	81	2.81 (0.93)	70	2	28

Note. Answer accuracy refers to the question "Choose the definition that seems most appropriate to you." Confidence in one's knowledge of the topic was rated on a scale from 1 to 5.

Table S6. Post-reading text ratings by text and version

	"Market"		"Social norms"		"Relief"		"Soil"		All texts	
Rating	Orig.	Mod.	Orig.	Mod.	Orig.	Mod.	Orig.	Mod.	Orig.	Mod.
Ease of following the text	2.77 (0.61)	2.86 (0.48)	2.47 (0.77)	2.95 (0.21)	2.43 (0.59)	2.47 (0.51)	2.50 (0.67)	2.35 (0.71)	2.55 (0.66)	2.66 (0.57)
Clarity	2.82 (0.39)	2.81 (0.40)	2.37 (0.83)	3.00 (0.00)	2.48 (0.59)	2.42 (0.61)	2.32 (0.72)	2.30 (0.70)	2.50 (0.66)	2.64 (0.57)
Wish to continue reading	2.45 (0.74)	2.52 (0.68)	2.47 (0.84)	2.41 (0.73)	2.00 (0.67)	2.00 (0.88)	2.00 (0.76)	1.91 (0.79)	2.22 (0.77)	2.21 (0.80)
Novelty	1.55 (0.67)	1.38 (0.50)	1.79 (0.79)	1.36 (0.58)	1.87 (0.46)	1.89 (0.66)	2.23 (0.53)	1.65 (0.57)	1.86 (0.65)	1.56 (0.61)

Note. Values are M (SD). Scale: 1 = “no”, 2 = “don’t know”, 3 = “yes”. Questions: ease of following — “Was it easy for you to follow the author’s line of thought?”; clarity — “Was the text clear to you?”; wish to continue reading — “In the textbook, this text continues. Would you like to continue reading?”; novelty — “Was there anything new for you in the text?”.

Table S7. Text comprehension by text and version

	“Market”		“Social norms”		“Relief”		“Soil”		All texts	
Measure	Orig.	Mod.	Orig.	Mod.	Orig.	Mod.	Orig.	Mod.	Orig.	Mod.
Number of correct answers (0–4)	2.59 (0.80)	2.52 (0.87)	2.37 (0.90)	3.09 (0.92)	2.26 (0.96)	1.95 (0.91)	1.82 (0.91)	2.35 (1.03)	2.26 (0.92)	2.49 (1.01)

Note. Values are M (SD).

Table S8. Percentage of correct answers to individual comprehension questions

	“Market”		“Social norms”		“Relief”		“Soil”	
Question	Orig.	Mod.	Orig.	Mod.	Orig.	Mod.	Orig.	Mod.
Q5	100	90	63	95	91	74	59	83
Q6	50	43	63	73	87	79	41	52
Q7	86	76	74	86	17	32	68	91
Q8	23	43	37	55	30	11	14	9

Note. Questions are numbered in the order of presentation after the text.

Effect of text version

Table S9. Effect of text version on eye-movement measures at the text level

	Time per character		Fixations per character		Fixation duration		Forward saccade amplitude		Regressive saccade amplitude	
Predictor	b (SE)	p	b (SE)	p	b (SE)	p	b (SE)	p	b (SE)	p
Intercept	4.07 (0.05)	< .001	−1.41 (0.04)	< .001	5.48 (0.02)	< .001	0.86 (0.03)	< .001	1.84 (0.04)	< .001
Version: modified	−0.08 (0.02)	< .001	−0.07 (0.02)	.001	−0.01 (0.01)	.19	0.01 (0.01)	.72	0.03 (0.02)	.17
Text: “Social norms”	−0.01 (0.03)	.78	−0.02 (0.03)	.51	0.01 (0.01)	.14	0.01 (0.02)	.68	−0.02 (0.03)	.53
Text: “Relief”	0.16 (0.03)	< .001	0.13 (0.03)	< .001	0.04 (0.01)	< .001	−0.06 (0.02)	.004	−0.04 (0.03)	.17
Text: “Soil”	0.09 (0.03)	.01	0.06 (0.03)	.07	0.03 (0.01)	< .001	−0.06 (0.02)	.003	−0.01 (0.03)	.85
Presentation order	−0.02 (0.01)	.02	−0.02 (0.01)	.02	−0.00 (0.00)	.95	0.01 (0.01)	.06	0.02 (0.01)	.01

Note. Models: log(measure) ~ version + text + order + (1 | participant). Reference levels: original version, text “Market”; presentation order was centered.

Table S10. Effect of text version on eye-movement measures in first paragraphs (contextualization)

	Time per character			Fixations per character		
Predictor	b (SE)	t	p	b (SE)	t	p
Intercept	4.07 (0.06)	68.24	< .001	−1.40 (0.05)	−27.50	< .001

	Time per character			Fixations per character		
Predictor	b (SE)	t	p	b (SE)	t	p
Version: modified	−0.13 (0.03)	−4.08	< .001	−0.13 (0.03)	−4.36	.001
Text: “Social norms”	0.13 (0.05)	2.79	.005	0.11 (0.04)	2.57	.01
Text: “Relief”	0.22 (0.05)	4.86	< .001	0.19 (0.04)	4.49	< .001
Text: “Soil”	0.14 (0.05)	3.03	.002	0.11 (0.04)	2.84	.005

Note. Models: log(measure) ~ version + text + (1 | participant). Reference levels: original version, text “Market”.

Table S11. Effect of text version on eye-movement measures in second and third paragraphs (dialogization and cohesion)

	Time per character			Fixations per character		
Predictor	b (SE)	t	p	b (SE)	t	p
Intercept	4.018 (0.057)	70.27	< .001	−1.462 (0.052)	−28.29	< .001
Version: modified	−0.090 (0.042)	−2.13	.033	−0.082 (0.041)	−2.01	.044
Technique: cohesion	−0.029 (0.042)	−0.70	.483	−0.027 (0.040)	−0.67	.502
Text: “Social norms”	−0.047 (0.043)	−1.10	.273	−0.058 (0.041)	−1.39	.163
Text: “Relief”	0.112 (0.043)	2.63	.009	0.081 (0.041)	1.99	.047
Text: “Soil”	0.107 (0.042)	2.58	.010	0.069 (0.040)	1.71	.087
Version × cohesion	0.047 (0.059)	0.79	.429	0.050 (0.057)	0.88	.381

Note. Models: log(measure) ~ version × technique + text + (1 | participant). Reference levels: original version, dialogization, text “Market”. The position of the techniques in the second and third paragraphs was counterbalanced using a Latin square.

Table S12. Effect of text version on reading of unchanged fragments

	Fragment reading time		Fixations on fragment		Fixation duration	
Predictor	b (SE)	p	b (SE)	p	b (SE)	p
Intercept	9.18 (0.13)	< .001	3.69 (0.13)	< .001	5.49 (0.02)	< .001
Version: modified	−0.14 (0.03)	< .001	−0.14 (0.03)	.001	−0.00 (0.01)	.73

Note. Models: log(measure) ~ version + (1 | participant) + (1 | fragment). Reference level: original version. Because the fragments are identical in both versions, the version effect on time and number of fixations per fragment equals the effect per character.

Table S13. Effect of text version on post-reading ratings

Rating	Original, M (SD)	Modified, M (SD)	b (SE)	t	p
Wish to continue reading	2.22 (0.77)	2.21 (0.80)	−0.009 (0.082)	−0.11	.912
Ease of following the text	2.55 (0.66)	2.66 (0.57)	0.108 (0.083)	1.30	.194
Clarity	2.50 (0.66)	2.64 (0.57)	0.130 (0.088)	1.47	.141

Rating	Original, M (SD)	Modified, M (SD)	b (SE)	t	p
Novelty	1.86 (0.65)	1.56 (0.61)	−0.289 (0.083)	−3.50	< .001

Note. Models: rating ~ version + text + (1 | participant). Reference levels: original version, text “Market”.

Table S14. Effect of text version on comprehension

Predictor	b (SE)	t	p
Intercept	2.451 (0.153)	15.98	< .001
Version: modified	0.211 (0.120)	1.76	.079
Text: “Social norms”	0.152 (0.173)	0.88	.379
Text: “Relief”	−0.448 (0.171)	−2.62	.009
Text: “Soil”	−0.471 (0.167)	−2.81	.005

Note. Model: comprehension ~ version + text + (1 | participant). Reference levels: original version, text “Market”.

Prior knowledge

Table S15. Prior knowledge and eye-movement measures at the text level

	Time per character			Fixations per character			Fixation duration		
Predictor	b (SE)	t	p	b (SE)	t	p	b (SE)	t	p
Between-person level									
Answer accuracy	0.03 (0.08)	0.39	.69	0.01 (0.06)	0.10	.92	0.02 (0.03)	0.86	.39
Confidence in the answer	0.01 (0.07)	0.16	.87	−0.01 (0.06)	−0.13	.90	0.02 (0.02)	0.77	.44
Familiarity with the topic	−0.05 (0.09)	−0.54	.59	−0.02 (0.07)	−0.21	.83	−0.03 (0.03)	−1.02	.31
Studied the topic at school	0.03 (0.08)	0.37	.71	0.05 (0.07)	0.72	.47	−0.02 (0.03)	−0.61	.54
Within-person level									
Answer accuracy	0.011 (0.016)	0.69	.492	0.009 (0.015)	0.62	.536	0.001 (0.003)	0.37	.711
Confidence in the answer	0.020 (0.018)	1.10	.272	0.021 (0.018)	1.17	.243	−0.001 (0.004)	−0.17	.864
Familiarity with the topic	0.039 (0.015)	2.59	.010	0.037 (0.015)	2.50	.013	0.002 (0.003)	0.66	.509
Studied the topic at school	0.015 (0.016)	0.98	.327	0.014 (0.015)	0.95	.342	0.001 (0.003)	0.25	.800

Note. A separate model was fitted for each prior-knowledge measure and each eye-movement measure: log(measure) ~ knowledge (within-person component) + knowledge (between-person component) + version + text + (1 | participant). Only the coefficients of the knowledge measures are shown. The interaction between version and prior knowledge was not significant in any model.

Table S16. Correlations of prior knowledge with post-reading ratings

	Wish to continue reading	Ease of following the text	Clarity	Novelty
Between-person level				

	Wish to continue reading	Ease of following the text	Clarity	Novelty
Answer accuracy	.29 (.060)	.34 (.025)	.32 (.034)	.07 (.637)
Confidence in the answer	.23 (.133)	.08 (.592)	.20 (.199)	.05 (.744)
Familiarity with the topic	.26 (.095)	.16 (.308)	.36 (.017)	−.18 (.240)
Studied the topic at school	.23 (.137)	.23 (.134)	.27 (.079)	−.12 (.448)
Within-person level				
Answer accuracy	.12 (.197)	−.03 (.731)	.16 (.079)	−.05 (.558)
Confidence in the answer	.12 (.209)	.16 (.083)	.15 (.093)	−.35 (< .001)
Familiarity with the topic	.19 (.033)	.18 (.045)	.22 (.014)	−.15 (.091)
Studied the topic at school	.01 (.886)	−.03 (.743)	.05 (.599)	−.03 (.711)

Note. Pearson correlation coefficients are shown as r (p). Between-person level: correlations of participant means. Within-person level: correlations of deviations from the participant mean and from the text mean; degrees of freedom were adjusted for the number of participants and texts.

Table S17. Correlations of prior knowledge with text comprehension

Prior-knowledge measure	Between-person level	Within-person level
Answer accuracy	.31 (.041)	.07 (.479)
Confidence in the answer	.26 (.094)	.01 (.894)
Familiarity with the topic	.18 (.250)	−.03 (.750)
Studied the topic at school	.12 (.440)	.08 (.386)

Note. Pearson correlation coefficients are shown as r (p). Between-person level: correlations of participant means. Within-person level: correlations of deviations from the participant mean and from the text mean; degrees of freedom were adjusted for the number of participants and texts.

Post-reading ratings

Table S18. Post-reading ratings and eye-movement measures at the text level

	Time per character			Fixations per character			Fixation duration		
Predictor	b (SE)	t	p	b (SE)	t	p	b (SE)	t	p
Between-person level									
Wish to continue reading	0.04 (0.06)	0.62	.53	0.02 (0.05)	0.47	.64	0.01 (0.02)	0.65	.52
Ease of following the text	−0.03 (0.08)	−0.42	.67	−0.06 (0.06)	−0.93	.35	0.03 (0.03)	1.01	.31
Clarity	0.06 (0.09)	0.69	.49	0.04 (0.07)	0.53	.60	0.02 (0.03)	0.69	.49
Novelty	0.06 (0.08)	0.82	.41	0.02 (0.06)	0.35	.73	0.04 (0.03)	1.52	.13
Within-person level									
Wish to continue reading	0.034 (0.02)	1.70	.09	0.04 (0.02)	1.86	.06	−0.00 (0.00)	−0.49	.62
Ease of following the text	−0.012 (0.02)	−0.76	.44	−0.01 (0.02)	−0.75	.45	−0.00 (0.00)	−0.14	.89

	Time per character			Fixations per character			Fixation duration		
Predictor	b (SE)	t	p	b (SE)	t	p	b (SE)	t	p
Clarity	−0.000 (0.02)	−0.03	.98	0.01 (0.01)	0.40	.69	−0.01 (0.00)	−1.85	.06
Novelty	0.009 (0.02)	0.52	.61	0.01 (0.02)	0.69	.49	−0.00 (0.00)	−0.67	.50

Note. A separate model was fitted for each rating and each eye-movement measure: log(measure) ~ rating (within-person component) + rating (between-person component) + version + text + (1 | participant). Only the coefficients of the ratings are shown.

Table S19. Correlations of post-reading ratings with eye-movement measures

	Time per character	Fixations per character	Fixation duration
Between-person level			
Wish to continue reading	.09 (.553)	.07 (.656)	.10 (.534)
Ease of following the text	−.07 (.655)	−.15 (.339)	.15 (.321)
Clarity	.10 (.531)	.07 (.638)	.10 (.512)
Novelty	.13 (.391)	.06 (.699)	.23 (.129)
Within-person level			
Wish to continue reading	.14 (.112)	.16 (.078)	−.05 (.599)
Ease of following the text	−.10 (.255)	−.10 (.272)	−.03 (.745)
Clarity	−.04 (.656)	−.00 (.995)	−.18 (.044)
Novelty	.13 (.144)	.14 (.120)	−.02 (.831)

Note. Pearson correlation coefficients are shown as r (p). Between-person level: correlations of participant means. Within-person level: correlations of deviations from the participant mean and from the text mean; degrees of freedom were adjusted for the number of participants and texts.

Table S20. Correlations of post-reading ratings with comprehension

Rating	Between-person level	Within-person level
Wish to continue reading	.12 (.459)	.10 (.302)
Ease of following the text	.08 (.615)	.03 (.711)
Clarity	.22 (.148)	.23 (.011)
Novelty	−.25 (.107)	−.06 (.501)

Note. Pearson correlation coefficients are shown as r (p). Between-person level: correlations of participant means. Within-person level: correlations of deviations from the participant mean and from the text mean; degrees of freedom were adjusted for the number of participants and texts.

Text comprehension

Table S21. Comprehension and eye-movement measures at the text level

	Within-person level			Between-person level		
Measure	b (SE)	t	p	b (SE)	t	p
Time per character	−0.004 (0.017)	−0.22	.828	−0.072 (0.067)	−1.08	.281
Fixations per character	−0.006 (0.017)	−0.34	.732	−0.056 (0.056)	−1.01	.313
Fixation duration	0.002 (0.004)	0.51	.611	−0.016 (0.024)	−0.66	.509

Note. Models: log(measure) ~ comprehension (within-person component) + comprehension (between-person component) + version + text + (1 | participant). Only the coefficients of comprehension are shown.